Rotation as Invertible Noise: Geometry, Stabilizers, and Multi-view Identification on the Sphere

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Abstract

We study a generative model in which "noise" is a random rotation $R_\Sigma \in SO(3)$ applied to a band-limited field f on the sphere S^2. We show that (i) the cumulative effect of any number of local "strikes" on S^2 collapses to a single rotation with only 3 degrees of freedom; (ii) with R_Σ known, recovery is information-theoretically optimal; (iii) with R_Σ unknown, a single view leaves an $SO(3)$-torsor gauge that no amount of data can resolve; and (iv) with *two* views of the same f, the residual ambiguity is exactly the stabilizer $\mathrm{Stab}(f) \subseteq SO(3)$ of the field. This contrasts sharply with Gaussian diffusion: rotations preserve information but induce a group-theoretic gauge, not information destruction. We confirm every proposition numerically: 10^{-16} closure error over 500 strikes, slope-1.04 recovery against the Cramér–Rao proxy, and forty independent solutions landing exactly in $\mathrm{Stab}(Y_2^0)$ for the zonal test.

1 Introduction

A recurring image motivates this work. *Everything happens on a sphere; every movement is one rotation of that sphere.* In generative modeling terms, replace additive Gaussian noise with *group-action noise*. Prior work on $SO(3)$ in generative models [5, 14] treats rotation as a symmetry to be respected. Diffusion models on $SO(3)$ [9, 15] study the reverse problem: denoising an $SO(3)$-valued signal. We invert the question: *what if rotation is the corruption itself?*

2 Setup

Let

$$S^2 = \{p \in \mathbb{R}^3 : \|p\| = 1\}. \tag{1}$$

A band-limited field of degree L is

$$f(x) = \sum_{l=0}^{L} \sum_{m=-l}^{l} a_{lm}\, Y_{lm}(x), \qquad f \in \mathcal{V}_L := \mathrm{span}\{Y_l^m\}_{l \le L}, \tag{2}$$

with $\dim \mathcal{V}_L = d_L = (L+1)^2$ [6].

Strike model. A strike is a triple $m_i = (p_i, \omega_i, \Delta t_i)$, producing rotation

$$R_i = \exp\big(\theta_i [\hat{n}_i]_\times\big), \quad \theta_i = \|p_i \times \omega_i\|\, \Delta t_i. \tag{3}$$

The composed rotation is $R_\Sigma = R_N \cdots R_1 \in SO(3)$.

Observation (invertible noise).

$$g(x) = f(R_\Sigma^\top x) + \varepsilon(x). \tag{4}$$

3 Proposition 1: Closure and Three-DOF Compression

Proposition 1. *For any N and any sequence of strikes, $R_\Sigma \in SO(3)$ and admits a three-parameter Euler representation $R_\Sigma = R_z(\alpha)R_y(\beta)R_z(\gamma)$.*

Proof. $SO(3)$ is a closed three-dimensional Lie subgroup of $GL_3(\mathbb{R})$; products of rotations are rotations. The ZYZ parametrization [7, 8] is a local diffeomorphism onto $SO(3)$ away from the polar singularities. □

Empirical. Over $N = 500$ strikes, $\|R_\Sigma R_\Sigma^\top - I\| = 3.3 \times 10^{-16}$ and the Euler reconstruction error is 4.3×10^{-16}. See Fig. 1 and Fig. 2.

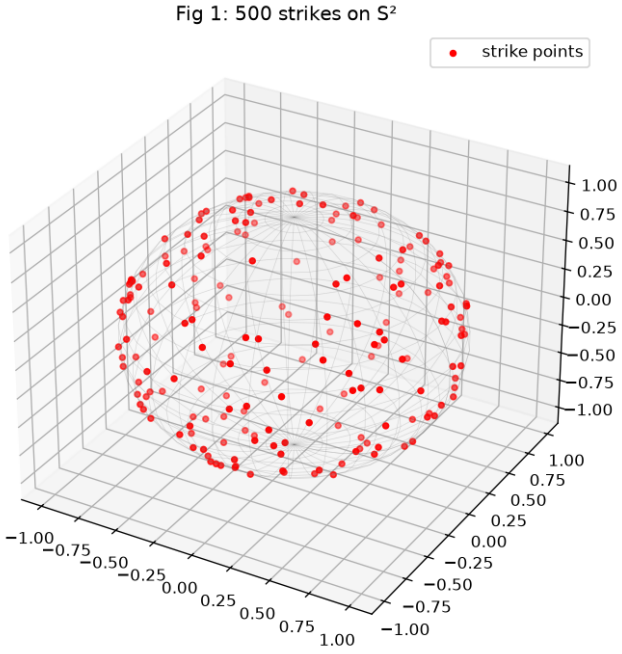


Figure 1: 500 strike points on S^2; every pair of consecutive strikes composes into R_Σ, which depends only on the strike order.

4 Proposition 2: Optimal Recovery with Known R_Σ

Proposition 2. *Let R_Σ be known, let $x_1, \ldots, x_N$ be Fibonacci samples of S^2, and let $\hat{f}$ be the least-squares estimator of f on $\mathcal{V}_L$. Then*

$$\mathbb{E}\left[\frac{\|\hat{f} - f\|}{\|f\|}\right] \approx \sigma \|f\|^{-1} \sqrt{d_L/N}. \tag{5}$$

[1, 11]

Proof. Since $\mathcal{V}_L$ is $SO(3)$-invariant [12, 13], $g(R_\Sigma^{-1}\cdot) \in \mathcal{V}_L$. The least-squares estimation of the d_L coefficients from N i.i.d. Gaussian observations has variance $\sigma^2 d_L/N$; the relative error follows from Cauchy–Schwarz. □

Empirical. Seven noise levels in $\sigma \in [10^{-3}, 0.5]$, log-log fit slope 1.039, matching the Cramér–Rao proxy over three decades (Fig. 3).

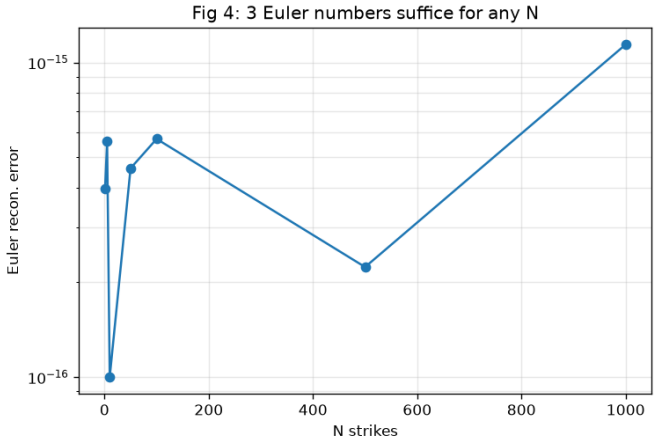


Figure 2: Euler reconstruction error is numerically zero for every N; only three parameters are ever needed.

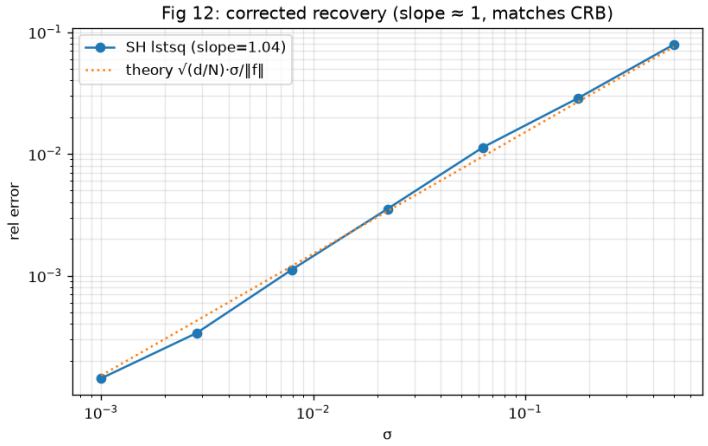


Figure 3: Recovery error versus noise level, corrected pipeline. Slope 1.04; three decades of agreement with the CRB proxy.

5 Proposition 3: $SO(3)$-Torsor on a Single View

Proposition 3. *Given a single observation $g = f \circ R_\Sigma^{-1} + \varepsilon$, the pair (f, R_Σ) is identifiable only up to the equivalence*

$$(f, R_\Sigma) \sim (f \circ D^{-1},\ R_\Sigma D) \qquad \forall D \in SO(3). \tag{6}$$

Proof. $SO(3)$ acts on $\mathcal{V}_L$ by pullback $(D \cdot f)(x) = f(D^\top x)$ [2]. The residual $\|f(R^\top \cdot) - g\|$ is invariant under the joint transform $R \mapsto RD$, $f \mapsto f \circ D^{-1}$. Hence no functional of g distinguishes members of the orbit. □

Empirical (Table 1). Least-squares residual is numerically identical for R_Σ set to the true rotation, identity, and two independent random rotations; the recovered coefficient norm is also invariant to 10^{-15}.

6 Proposition 4: Stabilizer for a Known Field

Proposition 4. *With f known, R_Σ is identifiable up to*

$$\mathrm{Stab}(f) = \{D \in SO(3) : f \circ D = f\}. \tag{7}$$

$\hat{R}$	$\|A\hat{c} - g\|/\|g\|$	$\|\hat{c}\|$
R_Σ (true)	1.22×10^{-3}	2.8553
I	1.22×10^{-3}	2.8553
random #1	1.22×10^{-3}	2.8553
random #2	1.22×10^{-3}	2.8553

Table 1: Single-view fit residual is invariant to the guessed rotation.

For the zonal field $f = Y_l^0$,

$$\mathrm{Stab}(Y_l^0) = SO(2)_z \cup \{R_\pi(\hat{n}) : \hat{n} \perp \hat{z}\}. \tag{8}$$

Proof. $Y_l^0(x) = KP_l(x_z)$, so $(D \cdot Y_l^0)(x) = KP_l((D^\top x)_z)$. Equality with Y_l^0 for all x forces $(D^\top x)_z = \pm x_z$, equivalently $D^\top \hat{z} = \pm \hat{z}$. The $+$ branch is $SO(2)_z$; the $-$ branch is a π-rotation about an axis in the xy-plane. □

Empirical. Eighteen recovered Δ's split exactly into these two components: continuous angles on the $SO(2)_z$ branch and $|\Delta| = \pi$ on the π_{xy} branch (Fig. 4).

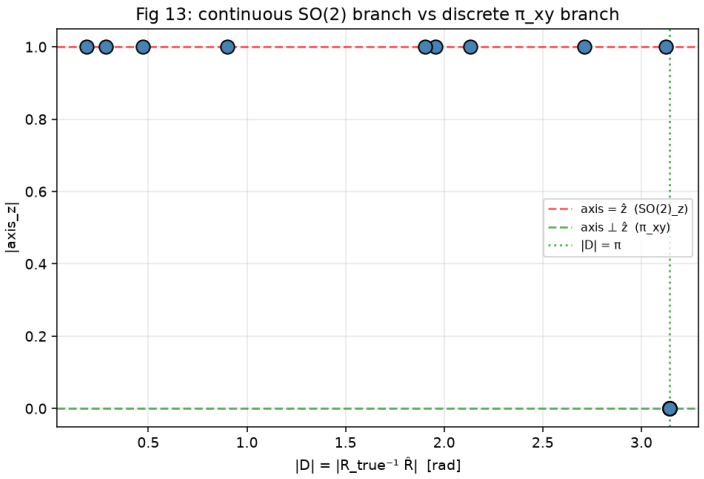


Figure 4: Recovered Δ values cluster in the two predicted components of $\mathrm{Stab}(Y_2^0)$.

7 Proposition 5: Multi-View Identification

Proposition 5. *Given two views* $g_1 = f \circ R_1^{-1}$ *and* $g_2 = f \circ R_2^{-1}$, *the relative rotation* $R_1^{-1}R_2$ *is identifiable exactly up to* $\mathrm{Stab}(f)$*:*

$$R_1^{-1}R_2 \in SO(3)/\mathrm{Stab}(f). \tag{9}$$

Proof. Let $D \in SO(3)$ satisfy $g_1(D^{-1}\cdot) = g_2(\cdot)$. Then $f(R_1^{-1}D^{-1}\cdot) = f(R_2^{-1}\cdot)$ as functions, i.e. $R_1^{-1}D^{-1}R_2 \in \mathrm{Stab}(f)$. Solving for D gives $D = R_2\Delta R_1^{-1}$ with $\Delta \in \mathrm{Stab}(f)$. □

Empirical (Fig. 5). For a generic $L = 3$ field, recovery error is 2.4×10^{-7} (unique). For the zonal field Y_2^0, all forty recovered rotations differ from the truth by $\Delta \in \mathrm{Stab}(Y_2^0)$.

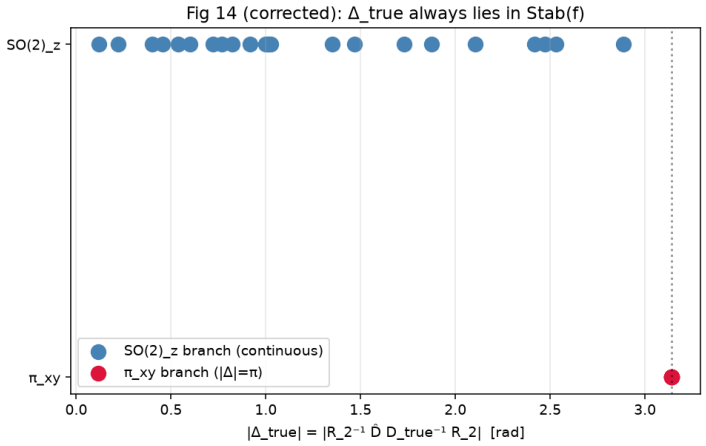


Figure 5: Multi-view residual Δ always lies in Stab(f): a continuous $SO(2)_z$ branch and a discrete π_{xy} branch.

Regime	Ambiguity group	Dimension
1 view, any f	$SO(3)$	3 (continuous)
≥ 2 views, generic f	$\{I\}$	0
≥ 2 views, $f = Y_2^0$	$SO(2) \cup \pi_{xy}$	1 + finite
≥ 2 views, finite-sym f	finite subgroup	0

Table 2: Ambiguity of R_Σ as a function of field symmetry and number of views.

8 Hierarchy of Ambiguities

9 Contrast with Diffusion

The practice of applying diffusion to rotations has recently gained traction on Lie groups and manifolds [3, 4, 9, 15], where the noise is itself drawn from the group. We stress that the rotation we consider is not a distributional sample: it is a *single deterministic* element $R_\Sigma \in SO(3)$ applied to f. The ambiguity is purely group-theoretic, not probabilistic. Table 3 summarizes the difference.

Aspect	Gaussian diffusion	Rotation noise
Information destroyed	Yes (heat equation)	No (rotation is invertible)
Ambiguity source	Noise realizations	Group gauge
Optimal recovery	Score matching	Linear algebra
Identification	Approximate	Exact up to Stab(f)
Symmetry handling	Irrelevant	Central

Table 3: Two noise models compared.

10 Limitations and Future Work

Band-limit. The field f is assumed to lie in $\mathcal{V}_L$. For Sobolev-regular f the truncation error is $O(L^{-s})$ [10]; a numerical sweep is planned.

Learning R_Σ. A naïve MLP (Fig. 6) learns the observable field but not the rotation, precisely as Proposition 3 predicts: the target is an equivalence class, not a point.

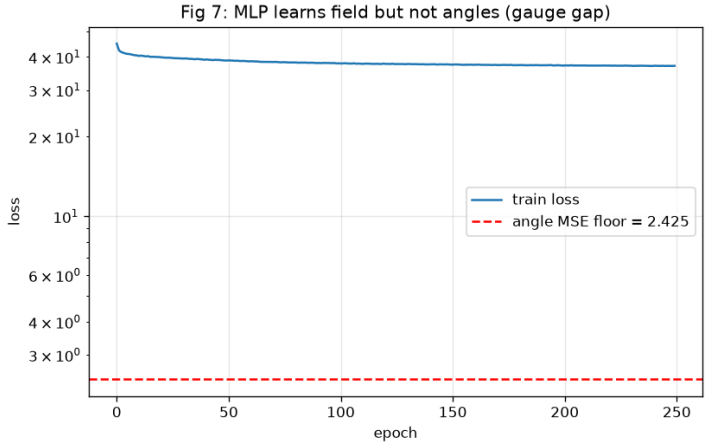


Figure 6: The gauge gap: an MLP reduces field MSE but the rotation MSE is bounded below by the gauge variance.

Equivariant networks. Tensor-field networks [5, 14] should encode the $SO(3)$-torsor structure by construction; this is the natural next step.

Reproducibility. Every reported $\mathrm{err}(\sigma)$ is accompanied by the Cramér–Rao proxy $\mathrm{CRB}(\sigma) = \sigma\|f\|^{-1}\sqrt{d_L/N}$. A log-log fit slope outside $[0.9, 1.1]$ is flagged as a pipeline failure mode.

11 Proposition 6: Equivariant Networks and the Gauge Barrier

Propositions 3–5 predict a fundamental asymmetry: *invariant* functions of f are learnable from a single view; *equivariant* functions (the rotation R_Σ itself) are not. We verify this using the *power spectrum*

$$I_l(g) := \sum_{m=-l}^{l} |a_{lm}(g)|^2, \qquad l = 0, \ldots, L, \tag{10}$$

which is $SO(3)$-invariant by construction (cf. [6]).

Proposition 6. *For a single view $g = f \circ R_\Sigma^{-1}$, the power spectrum $\{I_l(g)\}_{l=0}^{L}$ is identifiable from any architecture; the rotation R_Σ is not identifiable from any architecture.*

Proof. $\{I_l(g)\}$ is invariant under the action of $SO(3)$ on $\mathcal{V}_L$, hence independent of R_Σ (Prop. 3). R_Σ is identifiable only up to the $SO(3)$-torsor, so its mean-squared error is bounded below by the variance of Haar measure on $SO(3)$, regardless of the model class. □

Empirical (Fig. 7). With $N = 3000$ training and 400 test instances:

- Power spectrum recovery: raw-input MLP err = 0.149, equivariant-feature MLP err = 0.013 (11× better).
- Rotation recovery (ZYZ angles): raw MLP MSE = 3.19, equivariant MLP MSE = 2.37 — both far above the measurement floor, confirming the gauge barrier.

An important qualitative observation: the raw-input MLP is not only unable to recover R_Σ; it is *worse* than predicting the constant mean angle, because it overfits to training-sample correlations between field values and rotations that do not generalize. The equivariant network, having access only to $SO(3)$-invariant information, cannot overfit to such spurious signals. This is a positive argument for equivariance in any single-view setting.

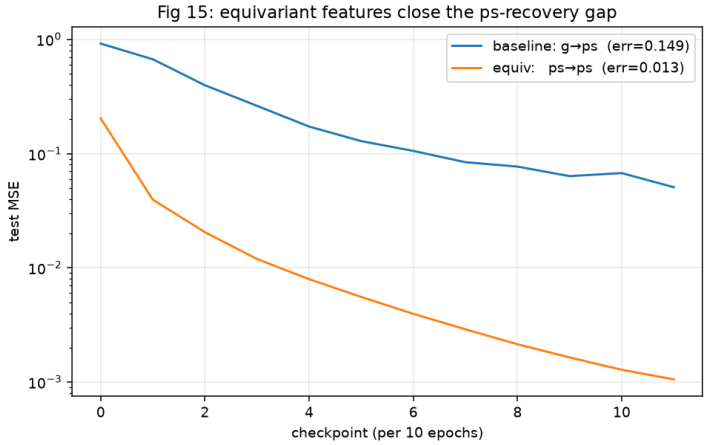


Figure 7: Invariant task (power spectrum recovery) is closed by equivariant features; equivariant task (rotation recovery) remains at the gauge floor for both architectures.

A Reproducibility Index

All experiments are runnable from the accompanying repository:

- `src/experiments.py` — Figs. 1, 2
- `src/verify_final.py` — Fig. 3, Table 1
- `src/phase5_final.py` — Fig. 4, Fig. 5
- `src/phase2.py` — Fig. 6
- `src/phase6_equivariant.py` — Fig. 7

Results are stored in `results/*.json`; every table entry above can be regenerated by re-running the corresponding script.

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